

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial – Midterm exam rehearsal - Solutions

Question 1

A factory grows yeast in 10,000 liters batch fermenter. After medium preparation and before its sterilization, the medium contains 10^5 spores per liter, while the sterilization demands are that the contamination probability will be 1 to 1000 fermentations.

The sterilization of the medium is carried continuously during the filling up of the fermenter, as the medium flows through a continuous sterilizer before entering and filling the fermenter (it can be assumed that the time it takes for the medium in the sterilizer to reach the sterilization temperature is very quick). The volume of the continuous sterilizer is 1,125 liters. The time it takes to fill the fermenter as described, is 80 minutes.

From professional literature you found out that sterilization of contaminating spores at temperature of 127°C for 11 minutes allows sterilization level of $1.85 \cdot 10^{-6}$, and that sterilization of these spores at temperature of 145°C for 12 minutes allows sterilization level of $3.2 \cdot 10^{-9}$.

The correlation between the sterilization constant (K_d) to the sterilization temperature is described by Arrhenius equation, meaning:

$$K_d = A \cdot e^{-E_a/RT}$$

A – empiric constant (min^{-1})

T – sterilization temperature ($^\circ\text{K}$)

E_a – activation energy ($\frac{\text{cal}}{\text{mole}}$)

R – universal gas constant ($1.98 \frac{\text{cal}}{\text{mole} \cdot ^\circ\text{K}}$)

What is the sterilization temperature to comply with the demand?

Solution:

Given:

<u>Literature</u>		<u>Fermentation</u>
$T = 127^{\circ}\text{C} = 400^{\circ}\text{K}$	$T = 145^{\circ}\text{C} = 418^{\circ}\text{K}$	$V_{ferm} = 10,000 \text{ L}$
$t_{st127^{\circ}\text{C}} = 11 \text{ min}$	$t_{st145^{\circ}\text{C}} = 12 \text{ min}$	$N'_0 = 10^5 \frac{\text{Spore}}{\text{L}}$
$\frac{N_f}{N_0} = 1.85 \cdot 10^{-6}$	$\frac{N_f}{N_0} = 3.2 \cdot 10^{-9}$	$N_f = \frac{1 \text{ Spore}}{1000 \text{ Ferm}}$
		$V_{sterilizer} = 1,125 \text{ L}$
		$t_{fill} = 80 \frac{\text{min}}{\text{Ferm}}$

 $T_{st} = ?$

Solution strategy:

- The sterilization temperature will be extracted from Arrhenius equation. We need to find K_d , A and Ea.
- Find the sterilization constant (K_d) in the given fermentation using the sterilization equation.
 - N_f is given in the question.
 - N_0 can be calculated from the fermenter volume.
 - Calculate the staying time from the medium flow rate and the sterilizer volume.
 - Calculate the medium flow rate from the medium volume and the time it takes to fill the fermenter.
- To find A and Ea we need the sterilization constants of the given temperature from the literature.
 - We will calculate the constants by the sterilization equation according to the sterilization period and the sterilization level described in the literature.
 - Write the two equations with two unknowns, isolate Ea first (by dividing the equations) and place to find A.
- Place K_d of the fermentation with the constant we found from the literature data.

Calculate the sterilization temperature from Arrhenius equation:

$$K_d = A \times e^{\frac{-E_a}{RT}}$$

First, calculate the sterilization constant that is required in the process described in the question, from the killing equation:

$$\ln\left(\frac{N_f}{N_0}\right) = -k_d \times t_{st}$$

N_f is given in the question. We will calculate N_0 according to N'_0 :

$$N_0 = N'_0 \times V_{ferm}$$

$$N_0 = 10^5 \frac{spore}{lit} \times 10,000 \frac{lit}{ferm} = 10^9 \frac{spore}{Ferm}$$

The staying time can be found from the medium flow rate through the sterilizer:

$$t_{st} = \frac{V_{sterilizer}}{Q}$$

While the medium flow rate can be found from the fermenter volume and the time it takes to fill it:

$$Q = \frac{V_{ferm}}{t_{fill}} = \frac{10,000 \frac{lit}{ferm}}{80 \frac{min}{ferm}} = 125 \frac{lit}{min}$$

Therefore, the staying time is:

$$t_{st} = \frac{V_{sterilizer}}{Q} = \frac{1125 \text{ lit}}{125 \frac{lit}{min}} = 9 \text{ min}$$

Hence, the sterilization constant in this fermentation is:

$$\ln\left(\frac{N_f}{N_0}\right) = -k_d \times t_{st}$$

$$\ln\left(\frac{0.001 \frac{spore}{ferm}}{10^9 \frac{spore}{ferm}}\right) = -k_d \times 9 \text{ min} \quad \rightarrow \quad k_d = 3.07 \frac{1}{min}$$

To find Ea and A - use literature data. First we will find the sterilization constant in each temperature:

$$k_d = \frac{\ln\left(\frac{N_f}{N_0}\right)}{-t_{st}}$$

$$k_{d_{418^\circ K}} = \frac{\ln(3.2 \cdot 10^{-9})}{-12 \text{ min}} = 1.63 \frac{1}{min}$$

$$k_{d_{400^\circ K}} = \frac{\ln(1.85 \cdot 10^{-6})}{-11 \text{ min}} = 1.2 \frac{1}{min}$$

Back to Arrhenius equation:

$$K_d = A \cdot e^{\frac{-Ea}{RT}}$$

$$1.63 \frac{1}{min} = A \cdot e^{\left(\frac{-Ea}{1.98 \frac{cal}{mole \cdot K} \cdot 418^\circ K}\right)}$$

$$1.2 \frac{1}{min} = A \cdot e^{\left(\frac{-Ea}{1.98 \frac{cal}{mole \cdot K} \cdot 400^\circ K}\right)}$$

Divide the equations to find Ea:

$$\frac{1.63 \frac{1}{min}}{1.2 \frac{1}{min}} = \frac{e^{\frac{-Ea}{827.64 \frac{cal}{mole}}}}{e^{\frac{-Ea}{792 \frac{cal}{mole}}}}$$

$$Ea = 5628.2 \frac{cal}{mole}$$

To find A, place Ea in one of the equations:

$$1.2 \frac{1}{\min} = A \cdot e^{\left(\frac{-5628.2 \frac{\text{cal}}{\text{mole}}}{1.98 \frac{\text{cal}}{\text{mole} \cdot \text{K}} \cdot 400^\circ \text{K}} \right)}$$

$$A = 1463.56 \frac{1}{\min}$$

Now, use Arrhenius equation with the killing constant required for the process described in the question, to find the sterilization temperature:

$$K_d = A \cdot e^{\frac{-E_a}{RT}}$$

$$3.07 \frac{1}{\min} = 1463.56 \frac{1}{\min} \cdot e^{\left(\frac{-5628.2 \frac{\text{cal}}{\text{mole}}}{1.98 \frac{\text{cal}}{\text{mole} \cdot \text{K}} \cdot T_{st}} \right)}$$

$$T_{st} = 460.93^\circ \text{K} \cong 461^\circ \text{K}$$

$$T_{st} = 188^\circ \text{C}$$

The sterilization should be carried in temperature of 188 degrees Celsius.

Question 2

A factory performs batch fermentation of yeast for the production of extracellular protein in 900-liter fermenter. The carbon source in the fermentation is glycerol and its initial concentration is 2%. The yeast is inoculated in initial concentration of 400 ppm and after 2 hours of lag, the yeast starts logarithmic growth. The yeast doubling time during logarithmic stage is 1.5 hour. At the end of the logarithmic growth, the fermenter is harvested, and its contents transferred to separation processes to produce the protein in a total yield of 86.9%. After discharging the fermenter, a process of cleaning, filling and sterilization is performed, so that when finished - the fermenter is ready for the next fermentation which immediately inoculated. The factory produces 168 kg of clean protein per month, while one working month includes 30 days and the factory works 24 hours per day.

Regular air is supplied into the fermenter at flow rate of 1500 LPM, temperature of 35°C and pressure of 1 atmosphere. It can be assumed that the air acts as ideal gas and that the air flow entering and exiting the fermenter are at the same rate.

Additional data: the oxygen solubility percentage at the end of the fermentation is 13%, the specific product production rate is $0.24 \frac{gP}{gX \cdot hr}$, the general biomass production rate is 1150 gram biomass per hour, the volumetric substrate consumption rate is $2.95 \frac{gS}{lit \cdot hr}$ and the glycerol is completely consumed during fermentation.

- A. What is the time period for cleaning and preparation in each fermentation cycle?**
B. What is the product yield on oxygen?

Solution:

Given:

<u>Fermentation</u>	<u>Yeast</u>	<u>Air</u>
$P_0 = 0 \frac{gP}{L}$	$X_0 = 400ppm = 0.4 \frac{gX}{L}$	$Q_{air} = 1500 \frac{L_{air}}{min}$
$S_0 = 2\% = 20 \frac{gS}{L}$	$t_d = 1.5hr$	$T = 35^\circ C = 308^\circ K$
$V = 900 L$	$q_P = 0.24 \frac{gP}{gX \times hr}$	$P = 1atm$
$\Delta t_{lag} = 2hr$	$R_X = 1150 \frac{gX}{hr}$	$\% Solubility_f = 13\%$
$\frac{W_{P_{clean}}}{month} = 168 \frac{KgP_{clean}}{month}$	$\frac{R_S}{V} = 2.95 \frac{gS}{L \times hr}$	
$\eta = 86.9\% = 0.869 \frac{KgP_{clean}}{KgP}$		
$S_f = 0 \frac{gS}{L}$		

- A. $t_{maintenance} = ?$**

Solution steps:

1. The maintenance time can be extracted from the length of each fermentation cycle. For that we need to find the duration of one fermentation cycle and that of the logarithmic growth.
2. The time period of one fermentation cycle can be calculated from the monthly output. For that we need to find the amount of product obtained in a single fermentation cycle.
3. The amount of product obtained in one fermentation cycle can be calculated from the production yield. For that we need to find the production yield and the biomass concentration at the end of the growth.
4. The biomass concentration at the end of the growth can be found from the growth yield.
5. The growth yield can be found from the general biomass production rate and the volumetric substrate consumption rate.
6. The production yield can be extracted from the specific protein production rate. For that we need to growth rate.
7. The growth rate can be found from the generation time.

The maintenance time is part of the fermentation cycle:

$$t_T = \Delta t_{lag} + \Delta t_{log} + \Delta t_{maintenance} \rightarrow \Delta t_{maintenance} = t_T - (\Delta t_{lag} + \Delta t_{log})$$

The protein production yield can be calculated by the specific production rate given in the question:

$$q_P = \mu \cdot Y_{\frac{P}{X}}$$

We need to growth rate – calculate it from the doubling time of the yeast in the fermentation:

$$\mu = \frac{\ln 2}{t_d}$$

$$\mu = \frac{\ln 2}{1.5 \text{ hr}} = 0.462 \frac{1}{\text{hr}}$$

Place the growth rate to find the protein production yield:

$$Y_{\frac{P}{X}} = \frac{q_P}{\mu}$$

$$Y_{\frac{P}{X}} = \frac{0.24 \frac{gP}{gX \cdot hr}}{0.462 \frac{1}{hr}} = 0.52 \frac{gP}{gX}$$

Find ΔP from the production yield:

$$Y_{\frac{P}{X}} = \frac{\Delta P}{\Delta X} = \frac{\Delta P}{X_f - X_0}$$

The yeast concentration at the beginning of the growth is given in the question. The yeast concentration at the end of the growth we can calculate from the growth yield:

$$Y_{\frac{X}{S}} = \frac{\Delta X}{\Delta S} = \frac{X_f - X_0}{S_0 - S_f}$$

The growth yield can be calculated from the general biomass production rate and the volumetric substrate consumption rate, given in the question:

$$Y_{\frac{X}{S}} = \frac{R_X}{\frac{R_S}{V} \cdot V}$$

$$Y_{\frac{X}{S}} = \frac{1150 \frac{gX}{hr}}{2.95 \frac{gS}{lit \cdot hr} \cdot 900 lit} = 0.43 \frac{gX}{gS}$$

Hence, the final biomass concentration is:

$$0.43 \frac{gX}{gS} = \frac{(X_f - 0.4) \frac{gX}{lit}}{(20 - 0) \frac{gS}{lit}}$$

$$X_f = 9 \frac{gX}{lit}$$

Place the yeast concentration at the end of the fermentation in the production yield:

$$\Delta P = Y_{\frac{P}{X}} \cdot (X_f - X_0)$$

$$\Delta P = 0.52 \frac{gP}{gX} \cdot (9 - 0.4) \frac{gX}{lit} = 4.47 \frac{gP}{lit}$$

It is given that the protein is extracellular, therefore:

$$\Delta P = (P_f - P_0) \rightarrow P_f = \Delta P + P_0$$

$$P_f = (4.47 + 0) \frac{gP}{lit} = 4.47 \frac{gP}{lit}$$

By multiplying the final concentration with the fermenter volume, we can find the protein amount that is produced in one fermentation:

$$\frac{W_{P_f}}{ferm} = P_f \cdot V$$

$$\frac{W_{P_f}}{ferm} = \frac{4.47 \frac{gP}{lit} \cdot 900 lit}{1000 \frac{KgP}{KgP}} = 4 \frac{KgP}{ferm}$$

This amount of protein is generated in one fermentation cycle, so by dividing the monthly protein amount in this amount - we can find the numbers of fermentation cycles done in a month, hence we can calculate the time period of one fermentation cycle.

Convert the amount of clean protein obtained in one month to the amount of protein produced per month:

$$\frac{W_P}{month} = \frac{\left(\frac{W_{P_{clean}}}{month} \right)}{\eta}$$

$$\frac{W_P}{month} = \frac{168 \frac{KgP_{clean}}{month}}{0.869 \frac{KgP_{clean}}{KgP}} = 193 \frac{KgP}{month}$$

From the monthly output of clean protein, calculate the amount of protein produces per month:

$$t_T = \frac{\frac{W_{P_f}}{ferm} \cdot \frac{hr}{day} \cdot \frac{day}{month}}{\frac{W_P}{month}}$$

$$t_T = \frac{4 \frac{KgP}{ferm} \cdot 24 \frac{hr}{day} \cdot 30 \frac{day}{month}}{193 \frac{KgP}{month}} = 15 \frac{hr}{ferm}$$

Use the logarithmic growth equation to find the period of logarithmic growth:

$$\ln\left(\frac{X_f}{X_0}\right) = \mu \cdot \Delta t_{log}$$

$$\Delta t_{log} = \frac{\ln\left(\frac{9 \frac{gX}{L}}{0.4 \frac{gX}{L}}\right)}{0.462 \frac{1}{hr}} = 6.7 \text{ hr}$$

Therefore, the maintenance time:

$$\Delta t_{maintenance} = t_T - (\Delta t_{lag} + \Delta t_{log})$$

$$\Delta t_{maintenance} = (15 - (2 + 6.7))hr$$

$$\Delta t_{maintenance} = 6.3hr$$

B. $Y_{\frac{P}{O}}$ = ?

Solution steps:

1. Calculate the product yield on oxygen by multiplication of the biomass yield on oxygen ($Y_{\frac{X}{O}}$) with the product yield ($Y_{\frac{P}{X}}$) that was calculated in section A.
2. The biomass yield on oxygen can be extracted from the maximal oxygen uptake rate.
3. Calculated the maximal oxygen uptake rate from the air flow rate and the oxygen solubility percentage at the end of the fermentation.

The biomass yield on oxygen can be calculate from the oxygen uptake rate:

$$O.U.R = \frac{\mu \cdot X}{Y_{\frac{X}{O}}}$$

The growth rate and the maximal biomass concentration (X_f) were calculated in section A.

The oxygen uptake rate we can calculate from the oxygen solubility:

$$O.U.R = \left(\frac{Q_{O_2}}{V} \right)_{in} \cdot (\%Solubility)$$

The question gives the solubility percentage at the end of the fermentation. In order to find the oxygen uptake rate, we need to convert the air flow rate to volumetric oxygen flow rate:

$$\begin{aligned} \left(\frac{Q_{O_2}}{V} \right)_{in} &= \frac{Q_{air}}{V} \cdot \%O_2 \cdot \frac{P}{T \cdot R} \cdot Mw_{O_2} \\ \left(\frac{Q_{O_2}}{V} \right)_{in} &= \frac{1500 \frac{lit_{air}}{min}}{900 lit} \cdot 60 \frac{min}{hr} \cdot 0.21 \frac{lit_{O_2}}{lit_{air}} \cdot \frac{1 atm}{308^\circ K \cdot 0.082 \frac{lit_{O_2} \cdot atm}{K \cdot mole O_2}} \cdot 32 \frac{g O_2}{mole O_2} \\ \left(\frac{Q_{O_2}}{V} \right)_{in} &= 26.61 \frac{g O_2}{lit \cdot hr} \end{aligned}$$

It is given that the solubility percentage at the end of the fermentation is 13%. This given solubility is the minimal required to enable enough oxygen to the biomass at the end of this specific fermentation. therefore, we can use it to calculate the oxygen uptake of the yeast at the end of the fermentation:

$$\begin{aligned} O.U.R_f &= \left(\frac{Q_{O_2}}{V} \right)_{in} \cdot (\%Solubility)_f \\ O.U.R_f &= 26.61 \frac{g O_2}{lit \cdot hr} \cdot \frac{13}{100} = 3.46 \frac{g O_2}{lit \cdot hr} \end{aligned}$$

The growth rate and the yeast concentration at the end of the fermentation were calculated in section A. Therefore, the biomass yield on oxygen:

$$O.U.R_f = \frac{\mu \cdot X_f}{Y_{X/O}}$$

$$Y_{\frac{X}{O}} = \frac{0.462 \frac{1}{hr} \cdot 9 \frac{gX}{lit}}{3.46 \frac{gO_2}{lit \cdot hr}} = 1.2 \frac{gX}{gO_2}$$

Hence, the product yield on the oxygen is:

$$Y_{\frac{P}{O}} = Y_{\frac{P}{X}} \times Y_{\frac{X}{O}} = 0.52 \frac{gP}{gX} \times 1.2 \frac{gX}{gO_2}$$

$$Y_{\frac{P}{O}} = 0.624 \frac{gP}{gO_2}$$